Genus-zero Gromov compactness
A proof blueprint for Lean formalization

Mathematical specification and agent task contracts

3 November 2026

Status. This document specifies a proof and its formalization obligations. It is not a kernel-checked proof. Analytic infrastructure is explicit work, not a collection of admitted axioms. "Ready" means that agents receive precise mathematical goals and dependencies; it does not mean that every task is a short tactic exercise. The estimates below are deliberately insensitive to numerical constants.

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1 Source audit and scope

The supplied paper is Max Lipyanskiy's Gromov–Uhlenbeck Compactness. Its main problem couples curves to anti-self-dual connections. Theorem 14 in Section 7.2, printed page 47, supplies a

curve compactness statement away from finitely many points. Section 5.8 contains a useful scalar concentration argument; Section 6.2 discusses elliptic bootstrapping. Section 8 concerns removal for matched pairs and refers elsewhere for interior curve singularities. These results do not themselves specify a stable sphere tree with energy identity. The gauge-theoretic constructions are unnecessary for our target theorem.

For the additional curve analysis, useful proof references are Aleksey Zinger's Notes on J-Holomorphic Maps: Proposition 4.1 (mean value), Corollary 4.2 (energy gap), Proposition 4.8 (regularity), Corollary 4.17 (cylinders), Proposition 5.1 (removal), Corollary 5.3 and Lemma 5.4 (concentration), and Proposition 5.5 (local trees). The following is a mathematical and software decomposition, with all intermediate contracts stated independently. No assertion is accepted merely by citing a paper.

We first prove sequential compactness for smooth spheres. Marked and already nodal inputs, integral homology preservation, and compactness of the quotient space are separately identified deliverables. No regularity, simplicity, somewhere-injectivity, genericity, semipositivity, or rationality of the symplectic form is assumed.

2 Exact statement and concrete definitions

2.1 Target, maps, energy

Fix a compact, boundaryless, finite-dimensional smooth manifold M, a smooth closed nondegenerate two-form ω, and a smooth endomorphism J of TM with $J^2 = -1$ and $\omega(v, Jv) > 0$ for $v \neq 0$. Set

$$g(v, w) = \tfrac{1}{2}\big(\omega(v, Jw) + \omega(w, Jv)\big).$$

This is a positive definite, J-invariant Riemannian metric. For compatible J it is the usual metric $\omega(\cdot, J\cdot)$. In distance arguments work on a connected component and let d be its Riemannian distance. A compact manifold has finitely many connected components; connected sphere images and a further subsequence reduce the general theorem to one of them. A curve means a smooth map satisfying $\mathrm{d}u \circ j = J \circ \mathrm{d}u$. For a measurable subset A of a Riemann surface define

$$E(u; A) = \tfrac{1}{2}\int_A |\mathrm{d}u|_g^2 \,\mathrm{dvol}, \qquad E(u) = E(u; \Sigma).$$

The domain metric is conformal to j. The energy measure is independent of that choice and, for a curve, equals $u^*\omega$. In particular, it is nonnegative, conformally invariant, and additive. Even for merely tame J the equality holds on J-complex tangent planes with the metric just defined. Do not assert that $\omega(v, Jw)$ itself is symmetric under a tameness assumption.

Use $\mathbb{CP}^1 = \mathbb{C} \cup \{\infty\}$ with its standard two-chart complex structure. Möbius maps are genuine biholomorphic automorphisms of this sphere. Define $D_r(a) = \{z : |z - a| < r\}$ and $A(r, R; a) = D_R(a) \setminus \overline{D_r(a)}$. Boundary circles have zero energy measure for smooth maps, so open/closed conventions do not change these energies.

2.2 Sphere trees

A labelled sphere tree consists of a nonempty finite tree $T = (V, E_T)$, curves $u_v : \mathbb{CP}^1 \to M$, and a point z_{vw} on component v for each oriented edge $v \to w$. For each v, the points z_{vw} are pairwise distinct. For each edge, require

$$u_v(z_{vw}) = u_w(z_{wv}).$$

It is stable when every constant component has at least three incident edges. With k markings, use k labels attached to distinct smooth points of components, disjoint from nodes, and count markings as special points for stability. Nonconstant components need no lower bound on their number of special points. The equivalence to finiteness of automorphisms is a separate lemma.

For implementation, use vertices $\mathrm{Fin}(N+1)$ and either a finite simple tree or a parent function with strictly decreasing index off the root. Node positions are indexed by incident edges, rather than by arbitrary pairs with meaningless default values. A nodal manifold quotient is unnecessary for the core analytic theorem: matching component data already encode the map.

2.3 Convergence, stated before the compactness theorem

For a sequence $u_n : \mathbb{CP}^1 \to M$, a compactness witness consists of a strictly increasing $\sigma : \mathbb{N} \to \mathbb{N}$, a sphere tree, and Möbius maps $\phi_{v,n} : \mathbb{CP}^1 \to \mathbb{CP}^1$ (component coordinates to the original domain), satisfying the following conditions.

1. $u_{\sigma(n)} \circ \phi_{v,n} \to u_v$ in C^∞ on compact subsets of $\mathbb{CP}^1 \setminus Z_v$, where $Z_v = \{z_{vw} : w \sim v\}$.
2. For $v \sim w$,
$$\phi_{v,n}^{-1} \circ \phi_{w,n} \longrightarrow z_{vw} \quad \text{locally uniformly on } \mathbb{CP}^1 \setminus \{z_{wv}\}.$$
For any two distinct vertices the analogous collapse follows using the first and last edges of their path. Prove this compatibility; do not assume that pointwise limits can simply be composed at an exceptional point.
3. $E(u_{\sigma(n)}) \to \sum_{v \in V} E(u_v)$.
4. In the annular coordinates supplied by the construction, every edge has vanishing neck image diameter. More precisely, with child coordinates affine in parent coordinates $a_n + \lambda_n z$, the neck predicate is the quantified condition in B03 below. Stabilization transports this predicate to surviving edges. This is useful strengthening data, not a consequence of energy identity alone for arbitrary smooth maps.

For marked sequences also require $\phi_{v_i,n}^{-1}(p_{i,\sigma(n)}) \to p_i$. The convergence relation itself takes the already selected sequence as input and quantifies over the frames, not over another subsequence. Thus existence of a convergent subsequence is kept distinct from convergence of a given sequence. Use chartwise convergence of every finite derivative to define C^∞_{loc}; the phrase "all derivatives" must not hide a separate subsequence for each order.

Theorem 1 (Core target)**.** *Let $u_n : \mathbb{CP}^1 \to M$ be nonconstant J-holomorphic maps and let $B \geq 0$ satisfy $E(u_n) \leq B$ for every n. There is a subsequence with a stable sphere tree limit in the sense above. There is a target-dependent $h > 0$ such that the number P of nonconstant components satisfies $Ph \leq B$. If V has more than one vertex and G components are constant, then $G \leq P - 2$, hence $|V| \leq 2P - 2$. A one-vertex nonconstant tree is also allowed.*

For arbitrary sphere sequences, first take either an infinite constant subsequence (compactness of M gives a constant limit), or an infinite nonconstant subsequence and apply the theorem. An unmarked constant sphere is *not* a stable map. Do not silently include it in the stable conclusion.

3 Proof, with the hard steps exposed

3.1 Uniform local analysis

Establish constants $\varepsilon_{\mathrm{reg}}, C > 0$, depending only on the fixed target, such that any curve on D_r satisfies

$$E(u; D_r) < \varepsilon_{\mathrm{reg}} \quad \Longrightarrow \quad \sup_{D_{r/2}} |\mathrm{d}u|^2 \leq C r^{-2} E(u; D_r). \tag{1}$$

The proof is a scalar mean-value argument applied to the energy density, after a differential inequality and a maximum selection argument. The flat domain rescaling must give the factor r^{-2}. Interior elliptic estimates upgrade derivative bounds to bounds on all derivatives. Arzelà–Ascoli and one countable diagonal selection then give C^∞_{loc} compactness wherever gradients are locally bounded.

Small total sphere energy implies, by a finite domain cover and (1), small image diameter. The image lies in a contractible target chart, where $\omega = \mathrm{d}\alpha$. Stokes gives $E(u) = 0$, so the sphere is constant. Thus there is $\varepsilon_{\mathrm{sph}} > 0$ such that every nonconstant sphere has energy at least $\varepsilon_{\mathrm{sph}}$. This proof uses no discreteness of symplectic periods.

Prove, independently of compactness, constants $\varepsilon_{\mathrm{neck}}, C_0, a > 0$ with: for a curve v on $[-L, L] \times S^1$, $E(v) < \varepsilon_{\mathrm{neck}}$ and $2 \le T < L$,

$$\begin{aligned} E(v; [-L+T, L-T] \times S^1) &\le C_0 e^{-aT} E(v), \\ \operatorname{diam} v([-L+T, L-T] \times S^1) &\le C_0 e^{-aT/2} \sqrt{E(v)}. \end{aligned} \tag{2}$$

One route first uses target monotonicity and (1) to put the interior of a small-energy cylinder into one target chart. In that chart the short-loop action/isoperimetric estimate bounds the energy of a central strip by the slice energies at its ends. Its derivative as the strip shrinks is minus those slice energies. Integrating the resulting scalar differential inequality gives exponential decay. Applying (1) to unit disks and integrating along a path gives the diameter estimate. Trimming the initial unit collars only changes C_0. Every constant is independent of L.

Finally prove removal of punctures. A finite-energy punctured disk becomes a half-infinite cylinder. Its tail energy tends to zero. Cylinder localization gives a unique limit point in the compact target. Decay gives $E(u; D_r) \le C r^{2\alpha}$ for some $\alpha > 0$ and then $|\mathrm{d}u(z)| \le C|z|^{\alpha-1}$. Choose $p > 2$ with $p(1-\alpha) < 2$. The extension is $W^{1,p}$, satisfies the equation weakly, and elliptic regularity makes it smooth. The weak extension needs a cutoff or boundary-term argument; continuity alone is not a regularity theorem.

Fix once and for all

$$0 < h \le \min(\varepsilon_{\mathrm{reg}}/2, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}/2). \tag{3}$$

Shrinking h later is harmless if every earlier lower bound is transported.

3.2 Concentration and a genuine defect measure

After taking a subsequence, the finite energy measures on the sphere converge weakly to a finite measure μ. Define $S = \{p : \mu(\{p\}) \ge \varepsilon_{\mathrm{reg}}\}$. Disjoint atoms give $|S|\varepsilon_{\mathrm{reg}} \le B$. If $p \notin S$, choose a small closed disk of μ-mass strictly below $\varepsilon_{\mathrm{reg}}$. The closed-set weak convergence inequality gives the same strict energy bound for the sequence eventually. Apply (1), elliptic estimates, and a diagonal subsequence. The resulting map on $\mathbb{CP}^1 \setminus S$ has finite energy and extends over S. On the complement, smooth convergence identifies the measures. Therefore

$$\mu = \mu_{u_0} + \sum_{p \in S} m_p \delta_p, \qquad \lim_n E(u_n) = E(u_0) + \sum_{p \in S} m_p. \tag{4}$$

This also rules out small residual atoms away from S. For any sufficiently small coordinate radius r isolating p, the boundary has μ-measure zero, so $E(u_n; D_r(p)) \to E(u_0; D_r(p)) + m_p$. On expanding plane domains the same argument is applied to measures extended by zero to the sphere, with ∞ a prescribed excluded point. One obtains a finite singular set in $\mathbb{C}$; removal extends the limit across infinity. No energy identity on the compactification is asserted before accounting for the possible atom at infinity.

3.3 The single-concentration problem

We now have curves $f_n : D_1 \to M$ with uniformly bounded energy, $f_n \to f$ smoothly off 0, f smooth on D_1, and

$$E(f_n; D_r) \longrightarrow E(f; D_r) + m \quad (0 < r < 1), \qquad m > 0.$$

For an actual defect the epsilon estimate gives $m \geq \varepsilon_{\text{reg}} \geq h$. Choose $a_n \in \overline{D_{1/2}}$ maximizing the Euclidean-domain gradient $|\mathrm{d}f_n|$. Then $a_n \to 0$ and the maximum tends to infinity. These claims follow because any bounded-gradient subsequence would make the defect zero, while gradients are bounded on compact sets away from 0.

By continuity of the radial energy function choose $\lambda_n > 0$ so that

$$E(f_n; D_{\lambda_n}(a_n)) = m - h/2. \tag{5}$$

Existence, not uniqueness, is needed. For each fixed small r the limiting energy of $D_r(a_n)$ is $m + E(f; D_r) > m - h/2$, whereas the energy at radius zero is zero. This also shows $\lambda_n \to 0$; all disks used eventually lie in D_1. Set $v_n(z) = f_n(a_n + \lambda_n z)$ on the inverse image of $D_{1/2}$. These domains eventually contain every compact subset of $\mathbb{C}$. There is no need to claim that the domains are nested.

3.4 Why no energy or distance survives in the neck

Fix $R > e^2$ and a small $\delta > 0$. By the defect identity and (5),

$$\lim_n E(f_n; A(\lambda_n, R\delta; a_n)) = h/2 + E(f; D_{R\delta}).$$

For δ small this is less than $\varepsilon_{\text{neck}}$. Apply (2) to this annulus in logarithmic coordinates, removing a collar of width $\log R$ from both ends. The resulting annulus is $A(R\lambda_n, \delta; a_n)$. For large n it is nonempty and

$$\begin{aligned} E(f_n; A(R\lambda_n, \delta; a_n)) &\leq CR^{-a}\varepsilon_{\text{neck}}, \\ \operatorname{diam} f_n(A(R\lambda_n, \delta; a_n)) &\leq CR^{-a/2}\sqrt{\varepsilon_{\text{neck}}}. \end{aligned} \tag{6}$$

Here n is chosen after δ, which is chosen after R. Thus both neck energy and diameter vanish in that order. In particular the rescaled maps capture all the mass m in the limit $R \to \infty$. This is the precise step that a bare bubble extraction and an energy bound do not supply.

3.5 Split the mass and prove constant components branch

Apply concentration compactness on $\mathbb{C}$ to v_n. After removal we obtain a sphere v and finitely many finite concentration points $q_1, \ldots, q_k$ with masses $m_i \geq h$. Equation (6) proves

$$m = E(v) + \sum_{i=1}^{k} m_i, \qquad f(0) = v(\infty). \tag{7}$$

For the equality of values, compare points on the two ends of the neck, take $n \to \infty$, then $\delta \to 0$, then $R \to \infty$, using removal at both ends. Energy equality alone is insufficient for this comparison.

Suppose v is constant. Since $m > 0$, at least one m_i is positive, so gradients of v_n are unbounded near some finite point. The maximizing choice of a_n implies $|\mathrm{d}v_n(0)| \geq |\mathrm{d}v_n(z)|$ on its domain. Hence 0 is itself a concentration point. Its mass is at most $m - h/2$, by (5). The mass identity (7) forces at least one other concentration point. Thus $k \geq 2$. Including the parent node ∞, this constant sphere has at least three distinct special points. This argument permits other atoms on $|z| = 1$; no assumption about that circle being a continuity set is needed for the estimate on the atom at zero.

3.6 Well-founded induction replaces "repeat bubbling"

If v is nonconstant, $E(v) \geq h$. If v is constant, there are at least two children and each has mass at least h. In either case, for every child,

$$0 < m_i \leq m - h, \qquad \lfloor m_i/h \rfloor < \lfloor m/h \rfloor .$$

Induct on the natural number $\lfloor m/h \rfloor$. Apply the same construction inside disjoint small coordinate disks around the q_i and attach the child roots at these points. All non-root bubbles have parent attachment at infinity. Compose affine rescalings along the rooted paths. Each stage has finitely many children, so the final tree and the number of subsequence refinements are finite. Record a strictly increasing index function at every refinement and reindex all previously constructed data. The local tree has total energy exactly m, matching nodes, pairwise distinct node positions, and stable constant vertices.

3.7 Global assembly and stabilization

Apply the local result to each defect in (4), using disjoint disks. Use stereographic coordinates obtained from global Möbius maps on the original sphere. Consequently all composed coordinate rescalings extend to global Möbius maps. The root has frame equal to the identity before stabilization. The local identities telescope to the global energy identity. Scale separation, distinct attaching points, and the inverse affine formulas prove the frame transition condition, including its inverse direction.

Only the original component can be constant and unstable. Delete a constant root of degree one, or suppress a constant root of degree two. In the latter case join its two neighbors at their old root attachments. Matching values persist because the deleted map is constant. At any now-forgotten smooth point there is no remaining energy atom; the epsilon estimate and bootstrapping upgrade convergence across that point. This upgrade is a required lemma, not an assumption about deleting vertices. A constant isolated root means total energy zero, impossible for the nonconstant sequence by the sphere gap. This proves Theorem 1.

For $|V| > 1$, every vertex has degree at least one and each ghost has degree at least three. The tree identity gives $3G + P \leq 2(P + G - 1)$, so $G \leq P - 2$. Summing the nonconstant energy lower bounds gives $Ph \leq B$. For a stable k-marked tree with at least two vertices the same argument gives $G \leq P + k - 2$. Handle one-vertex trees separately.

4 Lean architecture and acceptance rules

4.1 Use real definitions, then prove abstraction lemmas

The target structure must contain genuine smooth manifold, two-form, almost-complex, metric, and compatibility data. A curve must satisfy an actual differential equation. Energy must be an integral. Smooth convergence must be chartwise derivative convergence, and a tree must contain actual component maps. Do not declare any of these as arbitrary predicates and claim to have formalized geometry. Generic combinatorial lemmas may abstract these objects, provided their instantiation is a separate checked theorem.

Represent measurable energy first as an $\mathbb{R}_{\geq 0} \cup \{\infty\}$-valued measure, with a finiteness proof, and expose a real-valued finite-energy wrapper. Never use `toReal` before proving finiteness. Use a single strictly monotone index map for each subsequence and prove composition and preservation lemmas once. For limits use `Filter.Tendsto`; for iterated small-radius limits use $\mathcal{N}[>]0$. Prefer the

epsilon-quantified neck predicate in B03 over a deeply nested `lim` expression with hidden existence obligations.

Build in four layers: finite tree and sequence bookkeeping; local analytic estimates; concentration/rescaling; theorem assembly. There should be no import path from the local analysis back to the compactness theorem. A conditional assembly theorem taking proved local lemmas as explicit hypotheses can be checked early, but its name and documentation must say "conditional".

4.2 Library audit, not assumed availability

The workspace initially has no Lean project or mathlib revision. An installed Lean 4.33.1 binary is not a dependency pin. Current online mathlib documentation contains smooth manifold definitions, compactness of bounded finite measures on compact spaces, and a distributional Sobolev module. These are candidate starting points. Their presence does not establish that the required nonlinear elliptic estimates, $W^{1,p}$ regularity, Riemann-sphere geometry, or geometric measure arguments are implemented. F00 must check actual declarations and signatures in the chosen checkout. In particular the word "Sobolev" does not certify the needed $p > 2$ local PDE toolkit.

Every task below has one proposed owner file. Shared interfaces may change only through an explicit coordinated revision of their dependents. Several analytic tasks are substantial infrastructure packages; split them further after F00 if needed, preserving their output contract. No fixed wall-clock promise is made.

4.3 What counts as completion

For a proof task: its theorem uses the intended concrete definitions; the dependency closure builds with the pinned toolchain; it contains no `sorry`, `admit`, unproved project `axiom`, or unsound kernel bypass; and `#print axioms` contains no `sorryAx` or project assumption. Standard Lean logical axioms such as choice, propositional extensionality, and quotient soundness are acceptable. Check the statement for vacuous hypotheses and weakened conclusions as well as checking axioms. For definition tasks: provide constructors, the basic coercion/equivalence lemmas, and an explicit review of every analytic and geometric meaning. Source citations and informal sketches are never formal dependencies.

5 Agent task contracts

All tasks inherit Section 2 and the acceptance rules. File paths and theorem names are proposals until F00 freezes the project API. "Core" tasks establish the theorem; "extension" tasks extend its scope.

F00: Pin the project and audit the library

Dependencies: `none`. **Track:** core.
Proposed file: `GromovCompactness/LibraryAudit.lean`.

Output. A compatible pinned Lean/mathlib project, a successful build, and an inventory mapping each required concept to an actual declaration or a new obligation. Audit smooth forms and Stokes, metrics and distance, sphere charts and Möbius maps, finite measures and weak convergence, Arzelà–Ascoli, Sobolev/Morrey/elliptic estimates, finite trees, and singular homology. Do not edit global elan configuration. Record versions and imports in the repo. If analysis prerequisites are

missing, add explicit child tasks before marking their parent proof complete. Inability to download dependencies is a build limitation, not evidence that a mathematical theorem is absent.

F01: Target geometry and the equation

Dependencies: `F00`. **Track:** core.
Proposed file: `GromovCompactness/Geometry/Target.lean`.

Output. Concrete `ClosedTameTarget` and `JHolomorphicOn` definitions. Prove that the symmetrized metric above is positive and J-invariant, and that the equation in a complex coordinate is $u_t = J(u)u_s$. Supply finite uniformly controlled charts, metric norm comparisons, a positive uniform convexity/chart radius, and coordinate invariance. These uniform constants must depend only on the fixed target. Include the finite-connected-component reduction used for Riemannian distance.

F02: Sphere and conformal reparametrizations

Dependencies: `F00`. **Track:** core.
Proposed file: `GromovCompactness/Geometry/Sphere.lean`.

Output. A concrete compact metrizable smooth Riemann sphere with charts z and $1/z$. Implement affine maps $z \mapsto a + \lambda z$ for real $\lambda > 0$, inversion, and their generated Möbius automorphisms. Prove holomorphicity, inverse formulas, compactness, and that stereographic coordinate changes used in the proof extend globally. General classification of all automorphisms is unnecessary for constructing the witnesses.

F03: Energy measure and its algebra

Dependencies: `F01,F02,F04`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/Energy.lean`.

Output. For smooth maps define the nonnegative energy measure and a finite real-energy API. Prove additivity, monotonicity, conformal invariance, zero energy iff constant on connected domains, and $E(u; A) = \int_A u^*\omega$ for curves. Include zero mass of points and smooth boundary circles. Establish continuity of disk energy in radius and convergence of energy on compact measurable regions from C^1 convergence on a neighborhood. Prove moving-disk convergence by inner/outer disk sandwiching when the limiting boundary has zero mass.

F04: Smooth local convergence on varying domains

Dependencies: `F01,F02`. **Track:** core.
Proposed file: `GromovCompactness/Topology/SmoothLocal.lean`.

Output. Define `SmoothLocalConvergesOn` using uniform convergence of chartwise derivatives of every order on compact subsets. For varying domains require every compact subset of the limit domain to be contained eventually. Prove uniqueness, restriction, subsequence preservation, equation closure in C^1, and convergence after fixed smooth coordinate changes. Include the moving-point evaluation lemma under local uniform convergence.

F05: Subsequence and diagonal bookkeeping

Dependencies: `F00`. **Track:** core.
Proposed file: `GromovCompactness/Topology/Subsequence.lean`.

Output. A strictly increasing index-map type, identity and composition, and finite/countable refinement lemmas. For countably many nested subsequences construct a strictly increasing diagonal that is eventually a subsequence of every earlier one. Transport all finite data families by the same map. Do not claim an intersection of arbitrary infinite subsequences is infinite.

F06: Labelled trees and convergence data

Dependencies: `F01,F02,F03,F04,F05`. **Track:** core.
Proposed file: `GromovCompactness/Topology/SphereTree.lean`.

Output. Concrete finite rooted tree data, half-edge node positions, matching, component energy, componentwise stability, and the convergence witness in Section 2. Prove grafting preserves the tree property and node distinctness when children use pairwise distinct finite points and their parent node is infinity. Provide relabelling equivalences and finite path operations. The convergence predicate must not contain compactness as a field.

A01: Scalar mean-value inequality

Dependencies: `F00`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/MeanValue.lean`.

Contract. For $b \geq 0$ there are $\eta, C > 0$ such that if $e \geq 0$ is C^2 on a neighborhood of $\overline{D_r}$ and $\Delta e \geq -be^2$, then $\int_{D_r} e < \eta$ implies $\sup_{D_{r/2}} e \leq Cr^{-2} \int_{D_r} e$. Prove the scalar Laplacian average estimate first, maximize $(r - |z|)^2 e(z)$, and use a disk on which e is controlled by its central value. Handle $b = 0$ and $e \equiv 0$. The Laplacian sign convention is part of the contract.

A02: Energy-density differential inequality

Dependencies: `F01,F03`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/Bochner.lean`.

Contract. There is $b \geq 0$ depending only on the target such that the Euclidean-domain density $e = |\mathrm{d}u|^2/2$ of every curve on a disk satisfies $\Delta e \geq -be^2$. Derive this from the differentiated equation or a covariant Bochner identity, absorbing derivative terms by a quadratic inequality. Record the smoothness order used and the uniform target bounds. No domain-curvature term is present because the domain coordinate metric here is flat.

A03: Epsilon regularity

Dependencies: `A01,A02`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/EpsilonRegularity.lean`.

Contract. Prove (1), uniformly in $r > 0$, the curve, and the disk center. For an open disk apply the closed-disk estimate on slightly smaller disks and pass to the limit. Supply the version on a compact subset using a finite disk cover. Acceptance includes the correct squared derivative and the factor r^{-2}.

A04: Euclidean elliptic and Sobolev toolkit

Dependencies: `F00`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/EllipticToolkit.lean`.

Contract. For $1 < p < \infty$, $k \geq 0$, and $D_r \Subset D_R$, prove the interior estimate for $D = \partial_s + J_0\partial_t$, $\|f\|_{W^{k+1,p}(D_r)} \leq C(\|Df\|_{W^{k,p}(D_R)} + \|f\|_{L^p(D_R)})$, including its distributional version. Supply local multiplication/composition estimates, Morrey embedding for $p > 2$, and compact embeddings/Arzelà–Ascoli needed below. This is an infrastructure package: split it into checked prerequisites if F00 cannot locate these results. Do not substitute an L^2 Fourier Sobolev theorem for the required general-p statement.

A05: Nonlinear elliptic bootstrap

Dependencies: `A04,F01,F04`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/Bootstrap.lean`.

Contract. A continuous $W^{1,p}_{\mathrm{loc}}$ weak solution of the curve equation, $p > 2$, is smooth. Smooth solutions with a uniform first-derivative bound on a disk have uniform bounds for every derivative on smaller disks. Work in a target chart, freeze J, absorb the small principal-coefficient perturbation, localize, and iterate. The weak regularity proof must justify the first gain of a derivative (e.g. difference quotients); it cannot begin by assuming $W^{2,p}$. For bounds, specify dependence on chart margins, derivative bound, order, and derivatives of J.

A06: Compactness under local gradient bounds

Dependencies: `A05,F04,F05`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/LocalCompactness.lean`.

Contract. On eventually exhausting domains, a sequence of curves with uniform gradient bounds on each compact subset has a single subsequence converging in C^∞_{loc} to a curve. Use compactness of the target for values, uniform derivative bounds, a countable domain exhaustion, and a diagonal over derivative orders and charts. No energy bound is needed in this lemma once the stated derivative bounds are supplied.

A07: Positive energy of a nonconstant sphere

Dependencies: `A03,F01,F02,F03`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/EnergyGap.lean`.

Contract. Produce $\varepsilon_{\mathrm{sph}} > 0$ such that every nonconstant curve $u : \mathbb{CP}^1 \to M$ has $E(u) \geq \varepsilon_{\mathrm{sph}}$. Cover the round sphere by fixed coordinate disks; epsilon regularity gives a global gradient bound and hence image diameter $O(\sqrt{E(u)})$. For sufficiently small energy the image lies in one contractible chart. Prove the local primitive and Stokes step, then deduce zero energy. This task requires neither bubbling nor removal nor any rationality of $[\omega]$.

A08: Target monotonicity

Dependencies: `F01,F03,A04`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/Monotonicity.lean`.

Contract. There are $c, r_0 > 0$ such that if u is a nonconstant curve on a compact connected surface with boundary, $x \in u(\mathrm{int}\Sigma)$, $0 < r \le r_0$, and $u(\partial\Sigma) \cap B_r(x) = \varnothing$, then $E(u; u^{-1}(B_r(x))) \ge cr^2$. Prove the local isoperimetric/coarea differential inequality and pass through regular radii, including critical radii by approximation. Any use of coarea, Sard, local branch behavior, or unique continuation must be located in mathlib or made a checked prerequisite, never described merely as "standard". The hypothesis that x is actually in the image is essential.

A09: Localization of a small-energy cylinder

Dependencies: `A03,A08`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/CylinderLocalization.lean`.

Contract. For every $d_0 > 0$ there exists $\eta(d_0) > 0$, independent of $L > 1$, such that $E(v) < \eta(d_0)$ implies $\operatorname{diam} v([-L+1, L-1] \times S^1) < d_0$. First bound each interior loop diameter by $C\sqrt{E(v)}$. If the full image has large diameter, select a middle slice separated from two bounding slices; apply target monotonicity to the intervening cylinder. Make the ordered-slice selection explicit. Cover short cylinders by a bounded number of disks. Choose d_0 below a uniform target chart radius for A10.

A10: Energy and diameter decay on long cylinders

Dependencies: `A09,A03,F03`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/CylinderDecay.lean`.

Contract. Prove (2) for fixed $C_0, a, \varepsilon_{\mathrm{neck}} > 0$. After A09 put the trimmed image in an exact chart. Bound the absolute short-loop action by $C \int_{S^1} |v_\theta|^2$ (subtract the loop average and use Poincaré). Stokes expresses strip energy as the difference of its end actions. For $F(T)$ equal to central-strip energy obtain $F(T) \le -CF'(T)$ and integrate. Use A03 to obtain pointwise gradient decay from nearby strip energies, then integrate a circular segment and a longitudinal segment. Check $2 \le T < L$, endpoint collars, and independence from the length L.

A11: Continuous and Sobolev extension at a puncture

Dependencies: `A09,A10,A03,F03`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/PunctureExtension.lean`.

Contract. A finite-energy curve on $D_1 \setminus \{0\}$ has a unique continuous extension, and in one target chart near its limit satisfies $E(D_r) \le Cr^{2\alpha}$ and $|\mathrm{d}u(z)| \le C|z|^{\alpha-1}$ for some $\alpha > 0$. For A09 apply estimates to increasingly long finite cylinders, so arbitrary pairs of tail points lie in a common trimmed cylinder. For the power bound let the far end of A10 tend to infinity. Choose $p > 2$ with $p(1-\alpha) < 2$, show the extension belongs to $W^{1,p}$, and prove the curve equation distributionally across zero using cutoff estimates.

A12: Removal of singularities

Dependencies: `A11,A05,F02`. **Track:** core.
Proposed file: `GromovCompactness/Analysis/Removal.lean`.

Contract. The extension in A11 is smooth and J-holomorphic, is unique, and has the same energy. By inversion, every finite-energy curve on $\mathbb{C}$ extends to $\mathbb{CP}^1$. Also allow finitely many punctures by disjoint charts. There is no appeal to compactness of moduli spaces in this proof.

C01: Weak compactness of energy measures

Dependencies: `F03,F05`. **Track:** core.
Proposed file: `GromovCompactness/Compactness/Measures.lean`.

Contract. Finite nonnegative Borel measures on a compact metric space with uniformly bounded total mass have a weakly convergent subsequence, with convergent total mass. Prove closed-set upper bounds and convergence on sets whose boundary has limiting measure zero. This may be a wrapper around actual mathlib Prokhorov declarations after F00; identify their exact topological hypotheses. Include extension by zero from a measurable open subset.

C02: Finite concentration and smooth convergence

Dependencies: `C01,A03,A06,A12`. **Track:** core.
Proposed file: `GromovCompactness/Compactness/Concentration.lean`.

Contract. From a bounded-energy sequence on the sphere, or eventually exhausting its complement of a prescribed finite set S_0, extract a finite set $S \supset S_0$ and a smooth sphere map with smooth local convergence off S. Away from S_0, every actual defect has mass at least $\varepsilon_{\mathrm{reg}}$. Use closed disks and the closed-set inequality, not the open-set lower bound in the wrong direction. Bound $|S \setminus S_0|$ by $B/\varepsilon_{\mathrm{reg}}$. Include the no-atom corollary: at an eventually interior point with zero limiting energy atom, convergence upgrades across that point.

C03: Defect identity and moving centers

Dependencies: `C02,F03`. **Track:** core.
Proposed file: `GromovCompactness/Compactness/Defect.lean`.

Contract. For full-sphere input prove (4) as a measure identity, not merely an inequality. In a disk with one defect m at zero prove $E(f_n; D_r(a_n)) \to m + E(f; D_r)$ whenever $a_n \to 0$ and r is small enough to isolate the defect. Use zero limiting mass on boundary circles and sandwich moving disks. Prove positive defect implies diverging maximum gradient on $\overline{D_{1/2}}$, after restricting and scaling an isolating disk.

B01: Choose a common positive threshold

Dependencies: `A03,A07,A10`. **Track:** core.
Proposed file: `GromovCompactness/Bubbling/Threshold.lean`.

Contract. Construct a record with $h > 0$ satisfying (3), the relevant regularity constants, sphere gap, and cylinder estimates. Expose the comparison lemmas. Later inductions quantify over this fixed record; a new independent choice of h at each bubble is forbidden.

B02: Center and calibrate a rescaling

Dependencies: `C03,B01,F02`. **Track:** core.
Proposed file: `GromovCompactness/Bubbling/Calibration.lean`.

Contract. Under the single-concentration hypotheses of Section 3, produce $a_n \to 0$, $\lambda_n > 0$ tending to zero, and an eventual tail such that a_n maximizes the gradient on $\overline{D_{1/2}}$ and (5) holds. The rescaled domains are inverse images of $D_{1/2}$ and eventually contain every compact subset of $\mathbb{C}$. Use the intermediate value theorem and continuity of radial energy. Do not assert strict

radial monotonicity, uniqueness of the radius, or nested domains. Transport the maximum-gradient inequality to every point of each rescaled domain.

B03: Neck energy and diameter with explicit quantifiers

Dependencies: `B02,A10,C03`. **Track:** core.
Proposed file: `GromovCompactness/Bubbling/Neck.lean`.

Contract. For every $\eta > 0$ there is $R_0 > 1$ such that for every $R \geq R_0$ there is $\delta_0 > 0$ such that for every $0 < \delta < \delta_0$ there is N such that for every $n \geq N$, $R\lambda_n < \delta$, and both $E(f_n; A(R\lambda_n, \delta; a_n)) < \eta$ and $\operatorname{diam} f_n(A(R\lambda_n, \delta; a_n)) < \eta$. Prove this using the annulus with radii λ_n and $R\delta$, the exact residual energy $h/2 + E(f; D_{R\delta})$, and collar width $\log R$. Supply a version for closed subannuli or recover boundary values by continuity. This contract fixes the order of choices and does not exchange limits.

B04: One-level mass decomposition and node matching

Dependencies: `B03,C02,C03,A12`. **Track:** core.
Proposed file: `GromovCompactness/Bubbling/OneLevel.lean`.

Contract. Produce a sphere v, distinct finite points q_i, and positive masses $m_i \geq h$ such that $v_n \to v$ smoothly off the q_i and infinity, $m = E(v) + \sum_i m_i$, and $f(0) = v(\infty)$. For mass equality, first partition a fixed rescaled disk into regular pieces and small defect disks, then use B03 to let its radius grow. For matching, compare moving points on opposite neck boundaries using F04, B03, and removal. Never infer equality from Fatou's lower bound alone.

B05: A constant rescaled component has two children

Dependencies: `B04,B02`. **Track:** core.
Proposed file: `GromovCompactness/Bubbling/GhostBranching.lean`.

Contract. If the sphere v in B04 is constant, there are at least two finite concentration points. A positive total defect gives diverging gradients near some finite point. The maximizing center forces divergence at zero, hence an atom there. For radii $r < 1$ its mass is bounded above by $E(v_n; D_1) = m - h/2$. It cannot be the only atom since their total is m. Include distinctness from the parent point ∞ and derive stability of this ghost. Positivity of energies by itself does not prove this result.

B06: Energy rank strictly decreases

Dependencies: `B04,B05,B01`. **Track:** core.
Proposed file: `GromovCompactness/Bubbling/Rank.lean`.

Contract. Every child satisfies $m_i \leq m - h$ and $\lfloor m_i/h \rfloor_{\mathbb{N}} < \lfloor m/h \rfloor_{\mathbb{N}}$. Split into nonconstant v (its energy is at least h) and constant v (another child contributes at least h). State the arithmetic lemma for nonnegative real masses and $h > 0$ separately so it can be proved generically. No induction on a real-valued quantity without a well-founded relation.

B07: Resolve one defect into a finite rooted tree

Dependencies: `B06,F06,F05`. **Track:** core.
Proposed file: `GromovCompactness/Bubbling/LocalTree.lean`.

Contract. For the single-concentration input, construct a finite rooted sphere tree attached to the parent value $f(0)$, with total energy m, smooth component convergence, calibrated edge necks, and stable constant bubble vertices. Induct on the rank in B06. Isolate each child in a disjoint disk, apply the induction hypothesis to its rescaled sequence, compose all subsequence maps, and graft. Provide all local affine frame data as output. State the induction property uniformly over all input sequences with the given rank, so further subsequences remain admissible.

G01: Assemble global frames and the prestable limit

Dependencies: `B07,C03,F02,F06`. **Track:** core.
Proposed file: `GromovCompactness/Compactness/Assembly.lean`.

Contract. Construct the global prestable sphere tree from the base map and all defect trees, with total energy equal to $\lim E(u_n)$, matching nodes, and global Möbius frames. Prove neighboring-frame collapse in both directions directly from the affine formulas and scale separation. Prove collapse for longer paths using compact subsets avoiding the exceptional point and distinct node positions, rather than unrestricted composition of limits. The original root is the only possibly unstable constant vertex.

G02: Stabilize without weakening convergence

Dependencies: `G01,C02,F06`. **Track:** core.
Proposed file: `GromovCompactness/Compactness/Stabilize.lean`.

Contract. Remove a constant original root of degree one, or suppress one of degree two, preserving energy, matching, and all convergence conditions. At a deleted node that becomes smooth, identify zero limiting atom and apply C02 to upgrade convergence there. For a newly joined edge combine the two neck estimates and the shrinking connecting core on the constant component. An isolated constant root is reported as a separate case. No quotient-space compactness or stable reduction theorem is available as an assumption.

G03: Core theorem and constant alternative

Dependencies: `G02,A07`. **Track:** core.
Proposed file: `GromovCompactness/Compactness/GenusZero.lean`.

Contract. Prove Theorem 1 except the numerical component bound supplied by G04. Nonconstant inputs have energy at least h, ruling out an entirely constant limit. For arbitrary inputs choose an infinite constant subsequence or an infinite nonconstant subsequence. In the constant case use compactness of M and supply smooth convergence of constant maps. The public theorem must use the actual target and map structures and discharge every analytic-package hypothesis by proved lemmas.

G04: Component counts

Dependencies: `F06,B01`. **Track:** core.
Proposed file: `GromovCompactness/Topology/ComponentCount.lean`.

Contract. For a stable tree of total energy at most B, prove $Ph \le B$. If $|V| > 1$, prove $G \le P - 2$ for unmarked trees and $G \le P + k - 2$ for k-marked trees. For arbitrary sizes use the safe bound $|V| \le \max(1, 2\lfloor B/h \rfloor + k - 2)$, with the expression inside the maximum interpreted in integers, not truncated natural subtraction. Use the sum of degrees and handle the one-vertex case explicitly.

G05: Final core integration and audit

Dependencies: `G03,G04`. **Track:** core.
Proposed file: `GromovCompactness/Main.lean`.

Contract. Export the complete theorem with component bounds, build its full dependency closure, and record `#print axioms`. Audit all definitions against Section 2 and all hypotheses against Theorem 1. Verify that the dependency graph contains no reverse edge from analysis to compactness, and that no conditional abstract result is presented as the geometric theorem. This is the completion gate for the core track.

E01: Integral homology class is preserved

Dependencies: `G05,F01`. **Track:** extension.
Proposed file: `GromovCompactness/Topology/Homology.lean`.

Contract. Eventually $(u_{\sigma(n)})_*[S^2] = \sum_v (u_v)_*[S^2]$ in $H_2(M;\mathbb{Z})$. Choose small disjoint node disks. On cores use uniform convergence and unique short geodesics to homotope maps. Put necks and all caps inside contractible target balls using node matching and the diameter estimates. Cap and pinch to obtain an integral singular-chain or homotopy decomposition. Check orientations. Equality of symplectic areas cannot establish this statement, especially in the presence of torsion or a nontrivial kernel of pairing with $[\omega]$.

E02: Component stability equals finite automorphisms

Dependencies: `F06,A05`. **Track:** extension.
Proposed file: `GromovCompactness/Topology/Automorphisms.lean`.

Contract. Prove that a nonconstant sphere curve has finite group $\{\phi \in \mathrm{Aut}(\mathbb{CP}^1) : u \circ \phi = u\}$, and deduce that componentwise stability is equivalent to finiteness of the automorphism group of a finite marked tree. Establish local unique continuation/discreteness of fibers for nonconstant curves as prerequisites (e.g. by a proved similarity principle). Choose three distinct target values in the image; their finite fibers give a finite permutation representation with trivial kernel, since a sphere automorphism fixing three points is identity. Constant components with fewer than three special points have infinite stabilizer.

E03: Fixed marked smooth-sphere sequences

Dependencies: `G05`. **Track:** extension.
Proposed file: `GromovCompactness/Compactness/Marked.lean`.

Contract. For fixed k, sequences of stable k-marked smooth sphere maps of bounded energy have a subsequence converging to a stable k-marked tree, including convergence of all labels. If $E = 0$, require $k \geq 3$. Track marked points in the component frames. When points collide with each other or with nodes, insert constant components using ratios of differences of the finitely many point coordinates. Prove finite marked-configuration compactness on the sphere separately by normalization of triples and induction on label clusters. Preserve node distinctness and matching. Retain these zero-energy components until all markings have been assigned; only then stabilize. Merely taking limits of markings can put a label at a node and is not enough. Use G04 to bound the final number of components.

E04: Already nodal stable genus-zero inputs

Dependencies: `E03,G04`. **Track:** extension.
Proposed file: `GromovCompactness/Compactness/NodalInput.lean`.

Contract. For fixed k, every bounded-energy sequence of stable k-marked genus-zero maps has a stable Gromov-convergent subsequence. The component bound gives finitely many combinatorial types. Pass to one labelled input type, regard each half-node as an auxiliary marking, and apply E03 to its component sequences with simultaneous subsequences. Glue the limits at the labelled half-nodes, whose evaluations agree by marked convergence. Prove the glued graph is a tree and is stable, or explicitly stabilize. Use the definition of convergence for nodal inputs (component refinements with persistent original nodes), not a single global frame on a nonexistent smooth input sphere. No smoothing of arbitrary nodal maps is assumed.

E05: Topological compactness of the moduli space

Dependencies: `E04,E02,E01`. **Track:** extension.
Proposed file: `GromovCompactness/Topology/ModuliCompactness.lean`.

Contract. Define the quotient of stable marked maps of fixed class by component isomorphism, construct a metrizable topology inducing the specified Gromov convergence, and prove compactness from E04 and E01. Prove uniqueness up to isomorphism of a Gromov limit and the comparison of convergence witnesses; they are additional obligations. For energy-bounded moduli use continuity of energy for closedness. Do not infer topological compactness in an unspecified topology merely from a subsequence theorem. This package may require a separate detailed topology blueprint after the core theorem; it is not on the critical path to G05.

6 Agent protocol and stress cases

The generated JSON manifest records exact dependencies and contains the LaTeX body of each contract. Generated prompt packets include the global definitions, acceptance rules, and the task's own contract. The executable workflow is: freeze definitions; prove low-level analysis; prove concentration; prove calibration and neck control; perform natural-number induction; assemble and stabilize; audit the concrete theorem. Dispatch independent ready tasks only after their interfaces

are fixed. An orchestrator should never count a cited theorem or a compiled statement containing `sorry` as completed.

Before accepting the design, check these mathematical stress cases:

1. $M = \mathbb{C}$ with constant maps escaping to infinity shows why an unrestricted noncompact target is insufficient even with energy zero.
2. $u_n(z) = nz$ as maps $\mathbb{CP}^1 \to \mathbb{CP}^1$ shows why convergence on the original domain alone loses a nonconstant component.
3. A constant base with several concentration clusters requires ghost components. They cost zero energy and cannot be counted by energy alone.
4. Multiple covers must be admitted: no proof step may require injectivity.
5. A concentration point on the calibration circle must not break the ghost-branching proof or a boundary-measure claim.
6. Strict and non-strict inequalities at energy threshold values must be consistent; use the margin in (3).
7. A marked constant sequence with three colliding domain labels requires domain reparametrization, despite having no analytic bubbling.
8. The maps u_n can lie in varying homology classes. Fixed area is neither needed for extraction nor sufficient by itself to identify a class.

For a noncompact target the same fixed-J proof can be localized if all images lie in one fixed compact subset of a smooth ambient manifold; extract uniform geometry on a compact neighborhood. Bounded geometry alone does not prevent escape of constant maps. For $J_n \to J$ smoothly on a compact target, the analytic constants and equations need a uniform-family version before the same extraction argument applies. These are extensions of hypotheses, not silent interpretations of the core theorem.